Journal of Terramechanics 86 (2019) 15–29

Contents lists available at ScienceDirect

Journal of Terramechanics

journal homepage: www.elsevier.com/locate/jterra

Hybrid Soft Soil Tire Model (HSSTM). Part II: Tire-terrain interaction

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ARTICLE INFO

Article history:
Received 7 August 2019
Accepted 11 August 2019
Available online 30 August 2019

Keywords:
HSSTM
Contact model
Deformable terrain
Tire model
Brush model
Tire-terrain interaction

ABSTRACT

This is the second of three papers on the HSSTM. Part I introduced the tire structure model. Part III presents the tire model parameterization and validation. Part II describes the computational contact models for evaluating the tire/terrain interaction. Its main objective is to characterize the terrain in the normal and tangential directions during tire-terrain interaction. A contact search algorithm examines tire nodal points close to the ground and detects the penetration of the contact surfaces. Next, the contact interface algorithm applies the contact constraints to minimize or eliminate penetration. For deformable terrain, the algorithms are based on semi-empirical formulations and are implemented using a numerical approach in a 3D scheme. The normal and shear stress distribution simulation results for rigid wheel-soil contact are validated against experimental data. The contact interface algorithm for non-deformable terrains is implemented using a distributed 3D brush model enhanced with a transient module for dynamic manoeuvres.

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1. Introduction

The tire forces and moments generated during tire-terrain contact are influenced by tire deformation as well as terrain normal and shear deformations. In Part I of the paper (Sandu et al., 2019), a lumped mass discretized tire structure model was developed using Kelvin-Voigt elements. In this paper, the mathematical framework for characterizing the normal and shear behavior of different deformable terrains including soft soil, as well as the shear behavior of non-deformable terrains are discussed. The contact models detailed in this study are integral components of the hybrid soft soil tire model (HSSTM) package that was briefly introduced in other papers (Sandu et al., 2014; Taheri et al., 2013). Part III (Sandu et al.) presents the tire model parameterization procedure, as well as the validation of the HSSTM.

The contact model describes the tire-terrain interaction in two main steps. Initially, this interface searches for the nodal points that are close to the ground (contact search algorithm), and once the contact is detected, the algorithm applies the required contact condition (contact interface algorithm). Therefore, the contact simulation is initiated by detecting the penetration between the objects and then, by applying the contact constraints (such as contact forces), this penetration is reduced, minimized, or eliminated (Nakashima and Oida, 2004; Zavarise and De Lorenzis, 2009).

The mechanics of the tire-terrain interaction is fundamentally important in characterizing the performance of the tire on the terrain. In this regard, the mechanics of the tire-terrain interaction is divided into on-road and off-road case studies. For the on-road studies, the terrain is considered as a solid (non-deformable) surface with irregularities. Therefore, only terrain shear characteristics are incorporated. For this type of terrain, researchers have developed various high-fidelity models that can capture the tire response properties (Pacejka and Bakker, 1992; Oertel, 1997; Chae, 2006; Mastinu and Pairana, 1992). Additionally, multiple commercial tire models have been developed for integration with vehicle simulation software (Gipser, 2007; Oertel and Fandre, 1999; Gallrein and Backer, 2007). For the off-road case, where the terrain is assumed deformable, the complexity of the contact modelling problem increases (Wong, 2008).

Different terrains respond differently to similar loading inputs. The key factor for characterizing this behavior is to describe different terrain responses to the loading inputs using empirical relationships such as stress-strain relationships, pressure-sinkage relationships, and shear stress-shear displacement relationships (Taheri et al., 2013). The structure of natural terrains varies greatly, and, as a result, the terrain responses cannot be modeled using a single theory or method. The techniques for investigating the tire interaction with deformable terrain can be categorized into three

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https://doi.org/10.1016/j.jterra.2019.08.004
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Nomenclature

A	area
A_l	rigid area of a track link as a proportion of its nominal contact area
A_u	parameter characterizing terrain response to repetitive loading
a	half of the contact patch
b	smaller dimension of a rectangular plate or the radius of a circular plate; width
b_{ti}	tire width
b_{tr}	track width
c	soil cohesion
c_{tx}	tire tread longitudinal damping
$C_{F\alpha}$	cornering stiffness
C_{Fy}	lateral stiffness
CI	cone index
D	diameter
E	modulus of elasticity
F	function; thrust, tractive effort
F_x	longitudinal force
F_y	lateral force
F_z	vertical force
κ	slip
j	shear displacement
k	stiffness; resultant pressure–sinkage parameter
k_c, k_ϕ	pressure–sinkage parameters in the Bekker equation
$k_c', k_\phi', k_c'', k_\phi''$	pressure–sinkage parameters in the Reece equation
k_0	parameter characterizing terrain response to repetitive loading
$k_{p1}, k_{p2}, k_{z1}, k_{z2}$	pressure–sinkage parameters for snow cover
k_u	parameter characterizing terrain stiffness during the unloading–reloading cycle
k_{tx}	tire tread longitudinal stiffness
K	shear deformation parameter
K_1, K_2	parameters characterizing the shear stress–shear displacement relationship
l	length
L	perimeter; characteristic length for an ice layer
L_t	length of track in contact with terrain
L_α	lateral relaxation length
m, m_m	parameters characterizing the relation between the strength of the muskeg mat and that of the underlying peat
n	exponent of sinkage
n_x, n_y, n_z	unit vectors parallel to the global X, Y, and Z axes
p	pressure
p_{c0}, p_{c1}, p_{c2}	collapse pressures for an ice layer
p_{cr}	critical inflation pressure
p_g	ground pressure
p_i	tire inflation pressure
p_u	pressure at the beginning of unloading in a loading–un loading–reloading cycle
p_w	pressure–sinkage parameter for a snow cover
k_c, k_ϕ	resistance due to terrain compaction
r	radius
$s_{\max}$	maximum shear stress
t_z	terrain surface normal vector
T	tension
v_s	Stribeck relative velocity
v_r	tire longitudinal slip velocity
V_j	slip velocity
W	load, weight
z, z_o	sinkage
z_u	sinkage at the beginning of unloading in a loading–un loading–reloading cycle
z_w	pressure–sinkage parameter for a snow cover
α	tire slip angle
γ	density
δ_t	tire deflection
θ	angle
μ	Poisson's ratio; drawbar coefficient; coefficient of friction
σ	normal stress
λ_s	shear displacement along tire terrain interface
τ	shear stress
ϕ	angle of shearing resistance
ω	angular speed

main groups (Taheri et al., 2015): (1) empirical methods, (2) analytical methods, and (3) semi-empirical methods.

For empirical models, researchers measure terrain responses under conditions that are close to the vehicle operational condition, and then construct the empirical functions through curve-fitting and statistical analysis of the measurement data. Although the empirical models are good for simple mobility assessments, they cannot be extrapolated beyond the test data used to establish them. The physics-based models incorporate methods including computational physics and numerical analysis to construct an applied framework for the complex problem of soil-tire interaction. These models require high computational resources to handle their highly discretized structures. Additionally, most the models in this group (except for very detailed FEM models (Taheri et al., 2014), Discrete Element Models (DEM, or smooth hydroparticle models) use simplified parameterization procedures to evaluate the material properties and simulation input parameters. Consequently, correlating the physics-based models results with the experimental data is challenging. On the other hand, semi-empirical methods employ the empirical relations in addition to analytical approaches in order to achieve high fidelity results while keeping the computational effort and number of model variables low. Therefore, they are considered as good candidates for use in multibody and vehicle simulations.

In the following sections, the semi-empirical terrain module developed for HSSTM simulation package is introduced. The terrain module consists of contact search and contact interface algorithms. The contact interface algorithms for deformable and non-deformable terrains are handled differently due to the unique characteristics of each case. For deformable surfaces, underlying empirical equations for terrain normal and shear responses as well as repetitive loading are described. These equations were implemented in a three dimensional numerical platform that enables tire dynamic simulations. Using the empirical equations to describe terrain behavior results in more accurate force and moment estimations with less computational complexity. However, the simulations are limited to cases similar to the experimental set-up, and extrapolation beyond test conditions will result in higher estimation errors.

Next, tire force and moment generation on non-deformable surfaces was modeled using a distributed 3D brush theory. This model was augmented with a transient component to account for tire

relaxation length. The distributed bush model is a physics-based model that provides insight into mechanics of the tire contact patch including the sticking and sliding regions.

Tire contact interface models should be studied independent of the tire structure model to evaluate its accuracy. The effect of a deformable tire structure is the added dynamic compliance at the contact patch, in addition to the effect of the tire inflation pressure. This viscoelastic behavior causes inputs to the rim not to be instantly perceived at the contact patch. In simple words, the tire structure acts as a spring and damper between rim and tread elements. To evaluate the non-deformable contact model, tire performance on a relatively soft soil was evaluated. In this case, tire deformation is negligible when compared with the terrain deformation. Therefore, the tire was simplified as a rigid cylinder, and terrain stress and strain distributions were calculated and compared with experimental data. In the case of the rigid terrain, the tire viscoelastic behavior cannot be ignored. Consequently, the tire structural dynamics was accounted for using the relaxation length model that delays the wheel inputs to the brush contact model. Finally, the accuracy of the brush model that was parametrized using Flat-Trac data is discussed. The co-simulation of the validated terrain model discussed in this study with the developed tire structure model is presented in Part III (Sandu et al.).

2. Terrain model

The simulation environment data flow during tire-terrain interaction is shown in Fig. 1. The developed tire model has to be able to function in a complete vehicle simulation environment, and to provide forces and moments at various input conditions. At the beginning of each simulation timestep, the vehicle model provides the tire model with wheel kinetic and kinematic variables, including spindle position, orientation, and velocity components. The tire model uses this data in addition to the previous state of the tire/road contact point and road surface to estimate the deflection of the tire. Next, the tire model provides the wheel center data and the approximate position of the tire contact patch to the terrain model, in order to calculate the new tire/terrain contact point coordinates. The terrain model estimates the normal and the shear pressure distribution at the contact patch and updates the terrain profile based on the penetration of the tire into the terrain. The tire model uses the pressure distribution at the contact patch to calculate the resultant forces and moments (F&M). The calculated F&M are transformed from the coordinate system located at the center of the contact patch to the coordinate system located at the wheel center. The vehicle model receives the tire F&M data at the wheel center and solves the inertia equations for the wheel assembly, which results in the updated wheel kinetics and kinematics at the end of the timestep.

3. Contact analysis

During the tire interaction with the terrain, the contact algorithm searches for the nodal points that are close to the ground and when the contact is detected it applies the required contact condition. Therefore, the contact simulation is initiated by detecting the penetration between the objects; next, by applying the contact constraints (such as contact forces), this penetration is reduced, minimized, or eliminated.

The master-slave contact method was used for developing the tire-ground contact detection algorithm. During master-slave contact, surfaces of the two objects have contact with each other, and each node of the slave surface is checked for penetration into the elements of the master surface. The master nodes are considered the lumped masses and the slave surface is assumed to be the ground. The ground elevation model is constructed using the three dimensional regular grid road data (RGR), that has uniform grid step in X and Y directions, as shown in Fig. 2. The typical grid size used in simulations was 0.001 m. (The grid size in Fig. 2 is for illustration only.)

The contact search algorithm can be divided into two main methods: node-to-segment correspondence search and node-to-node proximity search. The node-to-node proximity search algorithm looks for the closest master node to a certain slave node, and it is very useful for contact surfaces with matching mesh. Node-to-segment is mostly used for large deformation contact problems that exist between surfaces with non-matching meshes (Zavarise and De Lorenzis, 2009). In this method, the proximity of the slave surface nodes to the master surface segments are examined, and, by preventing the penetration of nodes on the slave surface to corresponding master segments, the non-penetration conditions are enforced, as illustrated in Fig. 3.

After contact is properly detected, the contact conditions are applied upon penetration using the contact interface algorithm. The two major contact algorithms that are used extensively are the Lagrange Multipliers and the Penalty Method (Taheri et al., 2014). The penalty method calculates the contact forces relative to the contact penetration, and no additional degree of freedom is introduced into the system. The amount of penetration is a function of bulk modulus, volume, and surface area of the master segment. The penalty parameter is analogous to a spring stiffness that produces a force based on its deflection. Therefore, this method does not consider a zero gap assumption between the contacting surfaces.

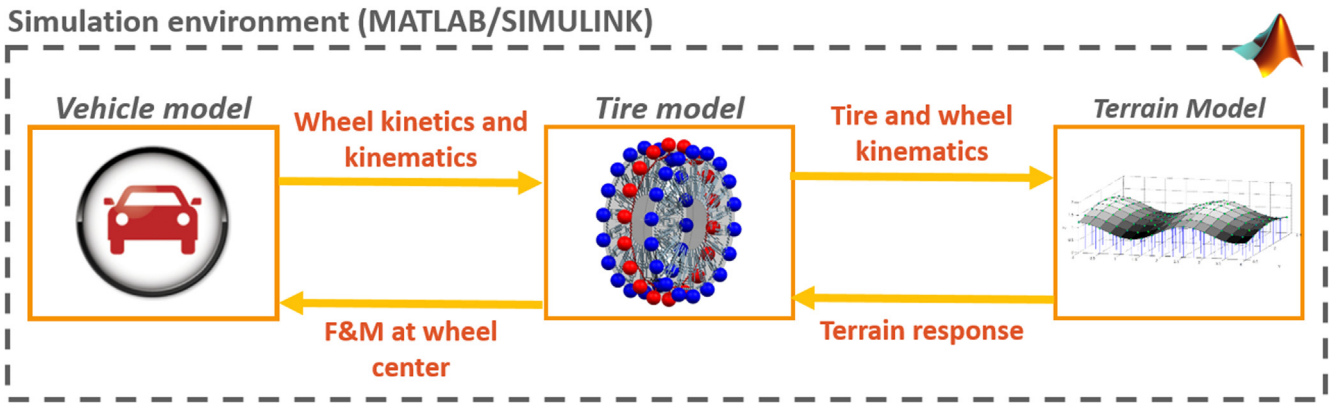


Fig. 1. Simulation environment data flow during tire-terrain interaction.

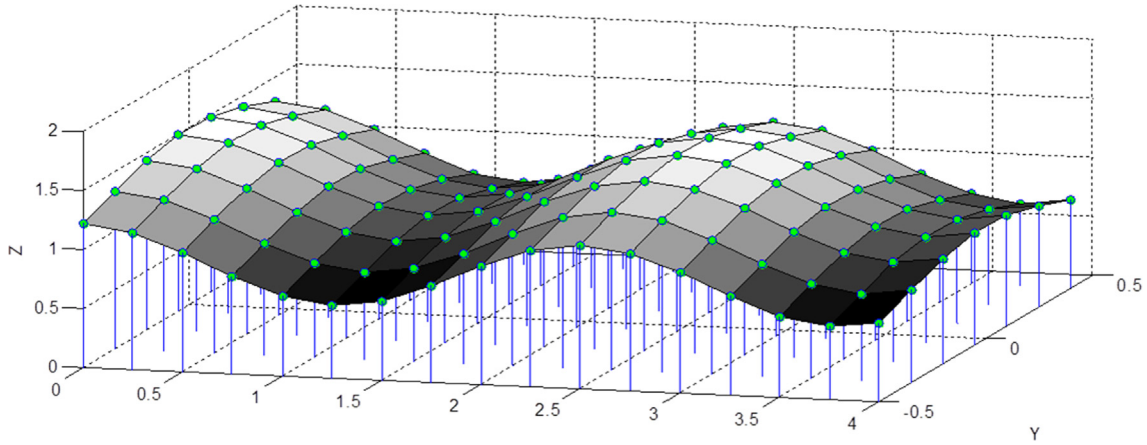


Fig. 2. Three dimensional soil elevation map representation using the three dimensional regular grid road (RGR) data files in SI units.

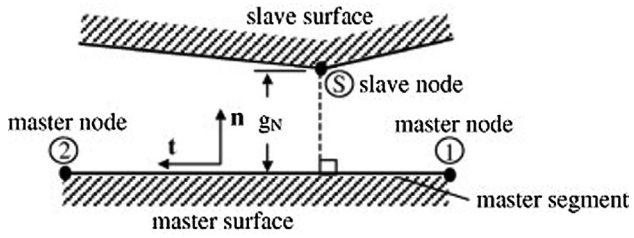


Fig. 3. Node-to-segment contact element geometry (used with permission of Zavarise and De Lorenzis, 2009).

The Lagrange Multiplier method adds a term, which consists of a Lagrange Multiplier coefficient and a gap function, to the energy equation of the system. This Lagrange Multiplier coefficient can be interpreted as a reaction force. It should be noted that the Lagrange Multiplier approach introduces new degrees of freedom into the system, which requires additional computational resources. Due to the extra complexity of the Lagrange Multiplier method, it is mostly suitable for problems with lower number of contact nodes. Therefore, the penalty contact method is used in this study for defining the contact between tire tread and road surface.

4. Tire interaction with deformable terrain

Once a tire contact node hits deformable ground, it causes the ground to deform, and, consequently, stress distribution appears at the contact patch. The generated stress field depends on the contact node velocity, angle of attack, and relative penetration into the ground in addition to the ground stiffness and damping. Different ground response scenarios that can happen based on the direction of the contact node velocity are shown in Fig. 4. In order to understand the physics of the tire-terrain interaction systematically and characterize it more accurately, the terrain responses to normal and shear loadings are studied separately. In the following sections, the methods that are used in this study for representing the terrain response to various types of loadings are described.

4.1. Terrain response to normal loading

When a tire is travelling over the terrain, normal pressure distribution is generated in the tire contact patch (normal to the tire-terrain interface plane). This pressure distribution increases the normal stress tensor in the soil layers that are closer to the surface. This makes the terrain surface deform accordingly, which causes the tire to sink into the ground. This behavior is represented using the pressure-sinkage relationships determined experimentally using a Bevameter. Different expressions have to be derived for different terrains, such as mineral, muskegs, and snow-covered terrains. In this study, the ground normal stress distribution formulation for mineral and snow covered terrains were implemented. The pressure-sinkage relationship for mineral terrains can be described using the equation initially proposed by Bekker (1969), Bekker (1960), Bekker (1956):

$$\sigma_n = (k_c/b + k_\phi)z^n = k_{eq} \cdot z^n \tag{1}$$

where p is the pressure distribution, b is the radius of the circular plate or the smaller dimension of a rectangular plate, z is the sinkage, and n, k_c, and k_ϕ are the pressure-sinkage parameters obtained empirically. The Bekker equation is limited to low velocity cases with low shear stress distribution. At high shear stress, soil shear deformation increases rapidly, which causes flow of the soil particles in the contact patch.

The terrain empirical equations were implemented using a numerical approach. At every timestep, the locations and the velocities of the tire tread elements are passed to the deformable terrain model. Using this information and the terrain geometry, which was updated as well at the previous timestep, the possibility of the contact is studied. In case of contact, the pressure distribution at the tire-terrain interface is calculated and is applied to the tire, as well as to the ground. At equilibrium, the deformation of the tire and the soil sinkage reach their steady-state values and do not change considerably.

The surface of the mineral terrains can be covered with vegetation including grass, sedges, mosses, and low woody shrubs. This type of terrain is called muskeg. The fragile surface mat can break down if the applied pressure exceeds a critical pressure. The sinkage corresponding to the critical pressure is called critical sinkage, beyond which the resistance to penetration suddenly decreases. The surface mat under normal load usually fails due the lower levels of tensile strength compare to the shear strength.

In modeling the muskeg terrains the surface layer is idealized as a membrane structure, and the peat beneath as a spring that produces resistive force proportional to the vertical deformation. The general pressure-sinkage relationship for this type of terrain is (Wong, 2010):

$$\sigma_n = kz_\circ + m_m z_\circ^2 (L/A) = kz_\circ + 4m_m z_\circ^2 (1/D_h) \tag{2}$$

where k is the peat stiffness, $z_\circ$ is the critical sinkage, A and L are the area and perimeter of the loading area, and D_h is the hydraulic

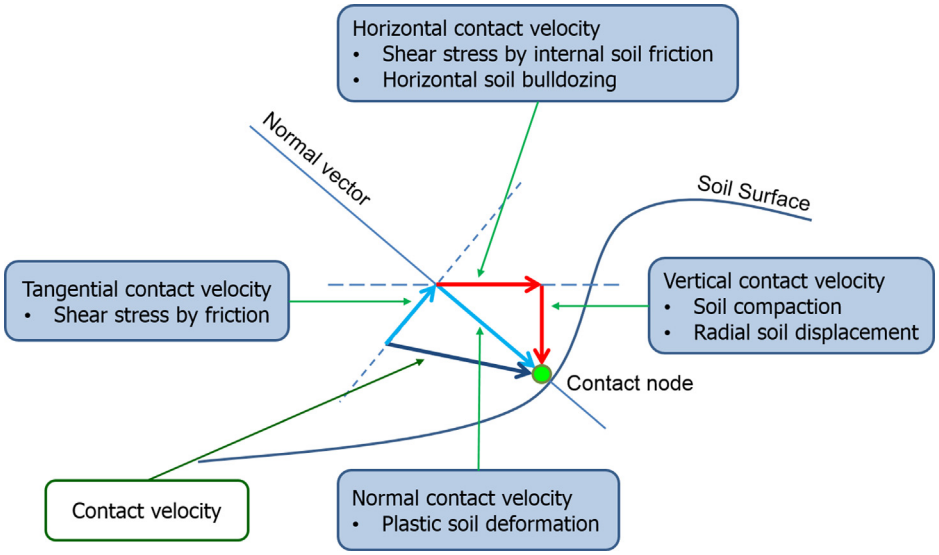


Fig. 4. Terrain responses to tire loading depend on contact node velocity, angle of attack, and relative penetration into the ground surface.

diameter of the plate and is equal to $4A/L$. The coefficient m_m is defined as:

$$m = \sqrt{kF_h/z_o^2} \tag{3}$$

With:

$$F_v/F_h = dz/dx \tag{4}$$

where F_v and F_h are the vertical and longitudinal components of the tension force that exists in the surface mat. The k, m_m, and $z_\circ$ are influenced by the rate of penetration, loading plate shape, and plate size, and should be specified accordingly.

The mechanical properties of the snow-covered roads are different as compared to the mineral terrains. For the snow-covered terrains, the pressure-sinkage characteristics change slightly once the ice layer below the surface of the snow fails. It has been shown that the exponential relationship can represent the shape of the pressure-sinkage curves during the terrain loading (Wong, 2008):

$$z = z_w[1 - \exp(-\sigma_n/\sigma_w)] \tag{5}$$

where p is the pressure distribution, z is the sinkage, z_w and σ_w are empirical parameters that depend on the structural properties of the snow. Therefore, different values of empirical parameters z_w and σ_w for before and after ice layer failure should be identified. The following equations are used for curve-fitting these empirical parameters (Wong, 2010):

$$\sigma_w = k_{p1} + rk_{p2} \tag{6}$$

$$z_w = k_{z1} + k_{z2}/r \tag{7}$$

where r is the radius of the loading plate, and k_{p1}, k_{p2}, k_{z1}, and k_{z2} are the shape coefficients that are functions of the snow geometrical (depth) and physical properties.

4.2. Terrain response to repetitive loading

When a wheeled vehicle travels over deformable terrain, the interface of the tire-terrain deflects. Once the leading wheel of the vehicle passes, the terrain is unloaded and, consequently, stresses are reduced. Due to the hysteresis, the force-deflection path is different for the loading and unloading phases. During reloading, the slope of the force-deflection curve is approximately similar to the one from the unloading step. The characteristics of the repetitive terrain loading, which is also known as multi-pass effect, for a mineral terrain is shown in Fig. 5.

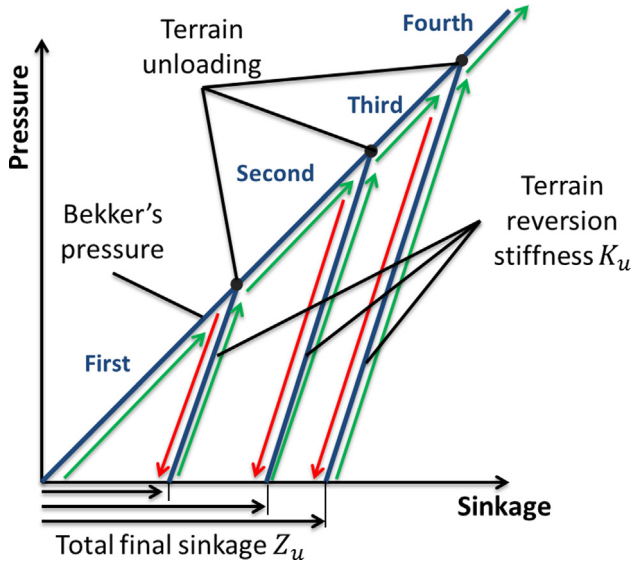


Fig. 5. Soil response to tire repetitive loading known as multi-pass effect (Wong, 2008).

The pressure-sinkage relationship for a mineral terrain during loading and unloading can be estimated using the following relations for the normal stress $\sigma_n$at loading (depending of the unloading parameters, denoted by subscript "u").

$$\sigma_n = \left(\frac{k_c}{b} + k_\phi\right) z_u^n - k_u(z_u - z) = k_{eq} z_u^n - k_u(z_u - z) = \sigma_u - k_u(z_u - z) \tag{8}$$

The terrain stiffness k_u during unloading is a function of unloading sinkage z_u:

$$k_u = k_\circ + A_u z_u \tag{9}$$

where $k_\circ$ and A_u are fitting parameters. This equation can be used for the repetitive loading of muskeg terrains when z_u is less than the critical sinkage z_b. At the end of unloading cycle $\sigma_n = 0$, and

elastic sinkage $z_e = z_u - z|_{\sigma_n=0}$. By substituting Eq. (9) in Eq. (10) terrain elastic sinkage z_e can be written as:

$$z_e = \left(\frac{\frac{k_c}{b} + k_\phi}{k_\circ + A_u z_u}\right) z_u^n \tag{10}$$

4.3. Soil damping

The damping behavior of the deformable terrain material during loading/unloading steps should be considered. The soil damping is particularly important when the tire is considered as rigid during the simulation. In this type of situation, due to the lack of structural damping from the tire, the transient tire-terrain interaction should be damped out by the terrain model in order to have overall model stability. The following formulation was used to model the damping force from the terrain (Mao and Han, 2008):

$$\sigma_{damping} = \begin{cases} -C_{ng} v_n^{i,j} & if \quad v_n^{i,j} < 0 \quad and \quad contact = true \\ 0 \end{cases} \tag{11}$$

where C_{ng} and $v_n^{i,j}$ are soil damping and velocity of the contact tire element in a normal direction relative to the soil surface. The soil damping can be found using the formulation suggested by McCullough (McCullough and Haug, 1986) as a function of soil shear deformation factor G, poison ratio υ, and tire contact area A_z:

$$C_{ng} = \frac{3.4GA_z^2}{1-\upsilon}\sqrt{G\gamma} \tag{12}$$

In order to find the tire element velocity component normal to the terrain surface, the terrain normal vector must be calculated. Consider the terrain at the immediate point of the tire tread element contact having the constant slopes of dZ/dX and dZ/dY in the global X and Y directions respectively. The ground normal vector at this point is derived as:

$$t_z = \frac{n_z - \left(\frac{dZ}{dX}\right)n_x - \left(\frac{dZ}{dY}\right)n_y}{\| n_z - \left(\frac{dZ}{dX}\right)n_x - \left(\frac{dZ}{dY}\right)n_y \|} \tag{13}$$

where n_x, n_y, and n_z are the unit vectors parallel to the global X, Y, and Z axes. The tire element velocity component normal to the ground surface is defined as $v^{i,j} \cdot t_z$.

4.4. Terrain response to shearing

The local shear behavior of the soil depends on the local slip velocity of the tire tread elements $\left(v_t^{i,j}\right)$, at the tire-terrain interface. The concept of the tire slip velocity on a deformable surface is similar to the one on a rigid surface. The slip velocity is the local tangential velocity of a tire tread element relative to the surface. In order to find this value the velocity vector of the tread element should be projected into the ground plane.

The projection of the tread element velocity $(v^{i,j})$ on the ground plane that has the normal vector of t_z is derived as:

$$v_t^{i,j} = v^{i,j} - \frac{v^{i,j} \cdot t_z}{\| t_z \cdot t_z \|^2} \cdot t_z \tag{14}$$

Then, the shear displacement along the tire-terrain interface λ_S is obtained by considering the tread element velocity for all the elements in the tire contact patch area:

$$\lambda_s = \int_{t_i}^{t_{i+1}} \sum_{i=1}^{N_{Belt_seg}} \sum_{j=1}^{N_{belt_elm}} v_t^{i,j} dt \tag{15}$$

Basically, this equation integrates the velocity of the individual bristles across the contact patch mesh at timestep t_i.

Previously, the pressure-sinkage characteristics during normal loading conditions were characterized for different types of loading curve shapes corresponding to different terrain types including mineral, muskeg, and snow-covered terrains. Similarly, these terrains exhibit unique shear stress-shear displacement behavior that needs to be formulated separately. For mineral terrains, the shear stress increases with shear displacement, and approaches a constant value. Using the Janosi and Hanamoto's approach (Janosi and Hanamoto, 1961), the shear stress distribution at the contact patch τ, is described as:

$$\tau/\tau_{\max} = \left(1 - e^{-\lambda_s/k}\right) \tag{16}$$

where $\tau_{\max}$ is the maximum shear stress, and K is the shear deformation parameter, which identifies the magnitude of the shear displacement required for the development of the maximum shear stress. This value is typically calculated from measurement data:

$$K = -\frac{\sum (1-\tau/\tau_{\max})^2 \lambda_s^2}{\sum (1-\tau/\tau_{\max})^2 \lambda_s [\ln(1-\tau/\tau_{\max})]} \tag{17}$$

The maximum shear stress $\tau_{\max}$ is defined using the Mohr-Coulomb equation:

$$\tau_{\max} = (c_a + \sigma_n \tan\phi) \tag{18}$$

where σ_n is the normal stress, ϕ is the angle of shearing resistance, and c_a is the soil cohesion (or adhesion). Similarly, a discretized version of shear stress equation is implemented in the model, which is used for integrating the stress field over the tire-ground contact patch and calculating the shear forces.

The shear curve of the muskeg terrains exhibit a maximum, and then the shear stress decreases continually with the increase of the shear displacement. This behavior is described as (Wong, 2008):

$$\tau/\tau_{\max} = (\lambda_s/K_w)\exp(1-\lambda_s/K_w) \tag{19}$$

The value of K_w is obtained by solving the following equation:

$$\sum (\tau/\tau_{\max})^2 [\ln(\tau/\tau_{\max}) - (1 + \ln(\lambda_s/K_w) - \lambda_s/K_w)][K_w - \lambda_s] = 0 \tag{20}$$

As for the snow-covered terrains, the shear curve displays a maximum and then decreases to an asymptotic value. In this case, the shear stress behavior is obtained as (Wong, 2008):

$$\tau/\tau_{\max} = K_r[1 + [1/K_r(1-1/e) - 1]\exp(1-\lambda_s/K_w)][1 - \exp(-\lambda_s/K_w)] \tag{21}$$

Using the Mohr-Coulomb equation is adequate for describing the maximum shear stress $\tau_{\max}$ in this equation.

5. Tire interaction with non-deformable terrain

The shear forces generated between the tire and the road have a significant role in the tire contact with a non-deformable (rigid) surface. Consequently, having a proper friction model is the key factor in identifying tire-road friction dynamics. The mechanics of the tire-road contact was described in this study using a brush model. This model consists of elastic bristles that generate forces proportional to their deflection in longitudinal and lateral directions. The tire F&M don't build up instantly upon actuating the wheel. The lag between tire inputs and F&M outputs was described using a relaxation length module (Rill, 2006). This module implicitly represents tire structure viscoelastic behavior, and can be used when a separate tire structure model is not used during simulation.

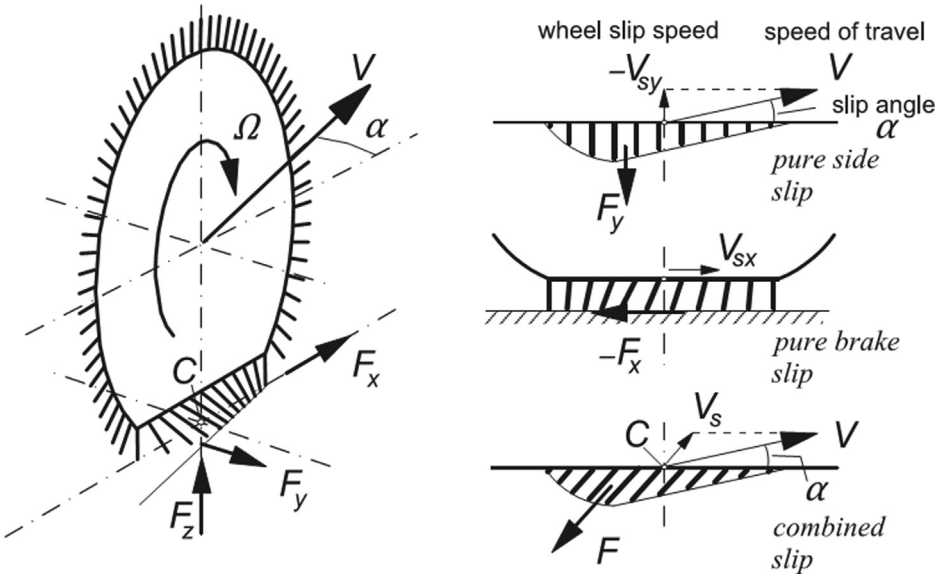


Fig. 6. Above the tire tread deflection is shown during longitudinal, lateral, and combined slip maneuvers (adapted after Rill, 2006).

5.1. 3D brush-type friction model

In order to model tire contact with non-deformable surfaces, the tire tread was modeled as brushes with stiffness in longitudinal and lateral directions, as seen in Fig. 6. The model is based on a 3D LuGre friction model that includes different friction effects including Coulomb or dry friction, viscous friction, Stribeck at low relative speeds, and stick in near zero speeds.

There are two representations for the LuGre model: the lumped version and the distributed version. Both versions are discussed next to explore their characteristics. The lumped LuGre model is formulated as:

$$F_x = (k_{tx}z + c_{tx}\dot{z} + c_v v_r)F_z \tag{22}$$

where z is the brush deformation, k_{tx} is the tire tread longitudinal stiffness, c_{tx} is the tire tread longitudinal damping, c_v is the viscous relative damping, and $v_r = r\omega - v$ is the relative velocity in the contact patch. The stiffness and damping values are measured per unit width of the tire. The brush deformation that is equivalent to tire tread deformation can be calculated using the following differential equation:

$$\dot{z} = v_r - \frac{k_{tx}|v_r|}{g(v_r)}z \tag{23}$$

With:

$$g(v_r) = \mu_c + (\mu_s - \mu_c)e^{-|v_r/v_s|^{\alpha}} \tag{24}$$

where μ_c is the normalized Coulomb friction, μ_s is the normalized static friction, is the Stribeck relative velocity, and α is a constant that is used to capture the steady-state friction force behavior.

For the distributed LuGre model the bristle deflection at a point$\lambda$of the contact patch is described using the following differential equation:

$$\frac{dz(\lambda,t)}{dt} = v_r(t) - \frac{k_{tx}|v_r|}{g(v_r)}z(\lambda,t) \tag{25}$$

The differential of bristle deflection can be written as:

$$dz(\lambda,t) = \frac{\partial z(\lambda,t)}{\partial \lambda}d\lambda + \frac{\partial z(\lambda,t)}{\partial t}dt \tag{26}$$

Rearranging this equation yields:

$$\frac{\partial z(\lambda,t)}{\partial t} = \frac{dz(\lambda,t)}{dt} - \frac{\partial z(\lambda,t)}{\partial \lambda}\frac{d\lambda}{dt} = \frac{dz(\lambda,t)}{dt} - r|\omega|\frac{\partial z(\lambda,t)}{\partial \lambda} \tag{27}$$

Substituting $\frac{\partial z(\lambda,t)}{\partial t}$ into the initial partial differential equation yields to:

$$\frac{\partial z(\lambda,t)}{\partial t} = v_r(t) - \frac{k_{tx}|v_r|}{g(v_r)}z(\lambda,t) - r|\omega|\frac{\partial z(\lambda,t)}{\partial \lambda} \tag{28}$$

For the combined slip situation where the tire slips relative to the ground in both longitudinal and lateral directions, the bristle deflection dynamic is captured using the following set of equations:

$$\begin{cases} \frac{\partial z_x(\lambda,t)}{\partial t} = v_{rx}(t) - \frac{k_{tx}|v_r|}{g(v_r)}z_x(\lambda,t) - r|\omega|\frac{\partial z_x(\lambda,t)}{\partial \lambda} \\ \frac{\partial z_y(\lambda,t)}{\partial t} = v_{ry}(t) - \frac{k_{ty}|v_r|}{g(v_r)}z_y(\lambda,t) - r|\omega|\frac{\partial z_y(\lambda,t)}{\partial \lambda} \end{cases} \tag{29}$$

These equations can be discretized in space for a finite number of bristles N:

$$\begin{cases} \dot{z}_{ix} = v_{rx}(t) - \frac{k_{tx}|v_r|}{g(v_r)}z_{ix}(\lambda,t) - r|\omega|\frac{N-1}{L}\left(z_{ix} - z_{(i-1)x}\right) \\ \dot{z}_{iy} = v_{ry}(t) - \frac{k_{tx}|v_r|}{g(v_r)}z_{iy}(\lambda,t) - r|\omega|\frac{N-1}{L}\left(z_{iy} - z_{(i-1)y}\right) \end{cases} \quad i = 2,\ldots,N \tag{30}$$

where z_{ix} and z_{iy} are the horizontal and lateral deflections of the ith bristle respectively, and L is the length of the contact patch. The developed distributed LuGre model is integrated with HSSTM tire model by connecting an array of deformable bristles to each of the tire belt elements. The array size (number of bristles in contact) is dynamic and depends on the normal load. Using this approach, assuming a predefined normal pressure distribution (such as parabolic) over the tire contact patch is not required, since the tire model can envelope over road obstacles and deform, thus producing a more accurate contact patch pressure distribution.

5.2. Relaxation length model

The steady-state F&M are considered time independent, which makes them functions of kinematic variables, such as slip angle or slip ratio. However, once these kinematic inputs are applied to the rolling tire, there is a time delay in the F&M generation. This dynamic lag, also known as the relaxation length, is defined as the distance the tire travels until the F&M reach steady-state, every

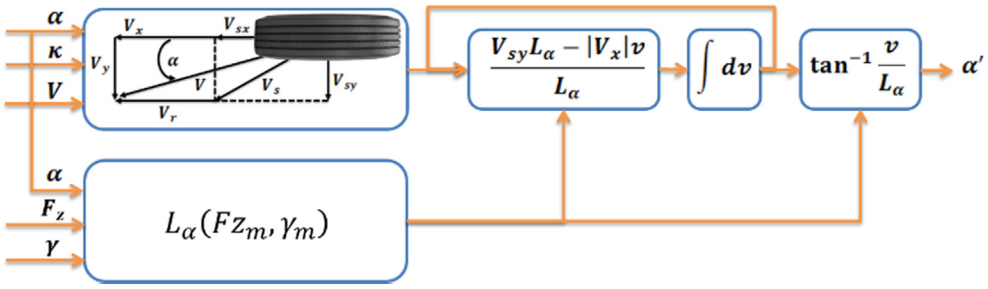


Fig. 7. The signal workflow for calculating the lagged slip angle.

time the input conditions change. The relaxation length is different for longitudinal and lateral tire dynamics. Usually, the relaxation length is higher for lateral forces due to the higher tire lateral structural compliance. For dynamic simulations, the solver frequency should be chosen greater than the highest studied frequency of the system in order to capture the delay properly, i.e., for lower relaxation length values the simulation timestep should become smaller.

In order to apply the transient effect to the brush model implemented, new lagged slip angle and slip ratio values are calculated and passed to the model (Pacejka, 2005). The lagged slip ratio and slip angles are input into the brush model to account for the tire structure dynamic compliance when the contact model is studied individually.

It is important to emphasize that this method was used in order to decouple the contact model from the tire structure model during the validation stage, while maintaining the time dependency of tire F&M. Consequently, please note that when the brush model was used in conjunction with the HSSTM tire structure model, the relaxation module was disabled.

In the lateral direction, the following differential equation was employed for estimating the tire lateral deflection due to side slip v_α:

$$\frac{dv_\alpha}{dt} + \frac{1}{L_\alpha}|V_x|v_\alpha = |V_x|\tan\alpha = -V_{sy} \tag{31}$$

where L_α is the lateral relaxation length, V_x is the longitudinal slip velocity, and V_{sy} is the lateral slip speed. Tire kinematic inputs are used to compute the velocity components in lateral and longitudinal directions, respectively:

$$V_x = V\cos\alpha \tag{32}$$

$$V_{sy} = |V_x|\tan\alpha \tag{33}$$

With the cornering slip stiffness $C_{F\alpha}$ and carcass lateral stiffness C_{Fy}, the lateral relaxation length may be obtained as:

$$L_\alpha = \frac{C_{F\alpha}}{C_{Fy}} \tag{34}$$

For linear, small slip conditions, the modified slip angle can be approximated as:

$$\tan\alpha' = \frac{v_\alpha}{L_\alpha} \tag{35}$$

The process workflow for calculating the lagged slip angle is shown in Fig. 7.

In order to find the relaxation length, Eq. (31) should be rearranged in terms of lateral force. The lateral force from the road surface to the tire at steady-state condition is:

$$F_y^{ss} = C_{F\alpha}\alpha = -C_{F\alpha}\frac{V_{sy}}{|V_x|} \tag{36}$$

Also, the lateral internal force due to carcass deflection that balances the lateral side slip force is:

$$F_y = C_{Fy}v \tag{37}$$

Substituting Eqs. (36) and (37) in (34) yields to the following equation:

$$L_\alpha\frac{dF_y}{dt} + |V_x|F_y = |V_x|F_y^{ss} \tag{38}$$

The left side of this differential equation represents the transient lateral force at the tire/terrain single contact point, and the right side corresponds to the lateral force at its steady-state condition $F_y^{ss}(\alpha, F_z, \gamma)$. This force is calculated directly from the pure slip lateral force regression function. The developed differential equation is equivalent to a first order delay:

$$\frac{F_Y}{F_Y^{ss}}(f_\alpha) = \frac{1}{i\left(2\pi\frac{L_\alpha}{V_x}f_\alpha\right)+1} = \frac{1}{\sqrt{\left(2\pi\frac{L_\alpha}{V_x}f_\alpha\right)^2+1}}\angle -\tan^{-1}\left(2\pi\frac{L_\alpha}{V_x}f_\alpha\right) \tag{39}$$

where f_α is the frequency of input slip angle changes. The effect of the slip angle frequency changes on the lateral force is shown in Fig. 8. At each frequency value, the slip angle increases from 0 to 1 degrees, then it decreases to −1 degree, and finally it increases again to 0 degrees. At very slow slip angle changes, the increasing and decreasing portions of the V_x vs. α curve are very close, but as the frequency increases, they separate. This is due to the fact that

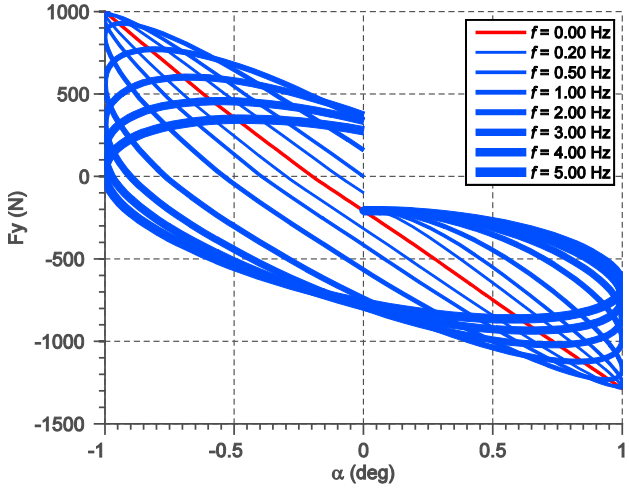


Fig. 8. The effect of the slip angle frequency changes on lateral force on a non-deformable surface.

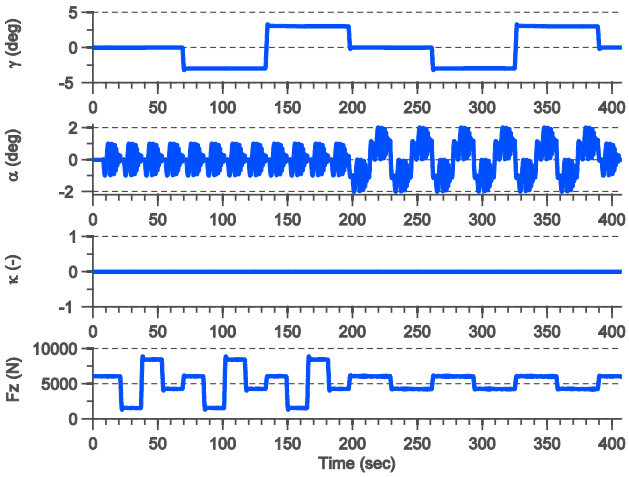


Fig. 9. F&M test input signals for transient cornering maneuver.

at high frequency inputs, the tire cannot respond quickly to its inputs, due to the hysteresis in its tread and sidewall.

By keeping the longitudinal slip at zero, the relaxation length can be calculated as:

$$L_\alpha = \frac{V_x}{2\pi f_\alpha}\sqrt{\left(\frac{F_Y^{ss}}{F_Y}\right)^2 - 1} \tag{40}$$

To estimate the relaxation length, a transient pure cornering experimental test was conducted on a Flat-Trac machine. For the test setup, the input slip angle was varied using the chirp input signal during multiple transient cornering cycles at various camber angles and normal loads. Meanwhile, the spindle was set to a neutral condition to keep the slip ratio zero. The time histories of the main input signals for the transient cornering test on 120 grid sand paper are shown in Fig. 9. These signals are the commanded signals that should be applied to the tire in order to study the lateral force behavior due to harmonic slip angles (α), and at different levels of constant inclination angle γ and normal load. The amplitude of slip angle changes are small in order to excite the tire in the linear range.

The recorded input and output signals including the lateral force from the spindle force hub were transformed to the tire contact patch coordinate system, and were divided into individual cycles for processing. Throughout each cycle, the slip angle input followed command chirp signal that sweeps multiple frequencies including 0.2, 0.5, 1, 2, 3, 4, and 5 Hz for the duration of 5, 2, 2, 1, 1, 1, and 1 s respectively. For calculating the amplitude ratio and phase difference between lateral force output and slip angle input, the corresponding signals were fitted to harmonic functions at every frequency value tested. Next, the amplitude ratio and phase difference terms in the first order delay Eq. (8) were fitted to the harmonic fit data points using the least-square method, and the lateral relaxation length values were calculated.

It should be pointed out that at the end of the described process, the relaxation length values were calculated at various normal loads, camber angles, and average slip angles. Using these data points, the regression function $L_\alpha(Fz_m, \gamma_m)$ was constructed. During the simulation at each timestep, the relaxation length was calculated using the function $L_\alpha(Fz_m, \gamma_m)$, and passed on to the differential equation (31) to find the lagged slip angle value, as seen in Fig. 7.

6. Results and discussion

The mathematical formulations for modeling the terrain behavior on non-deformable and deformable terrains were described in

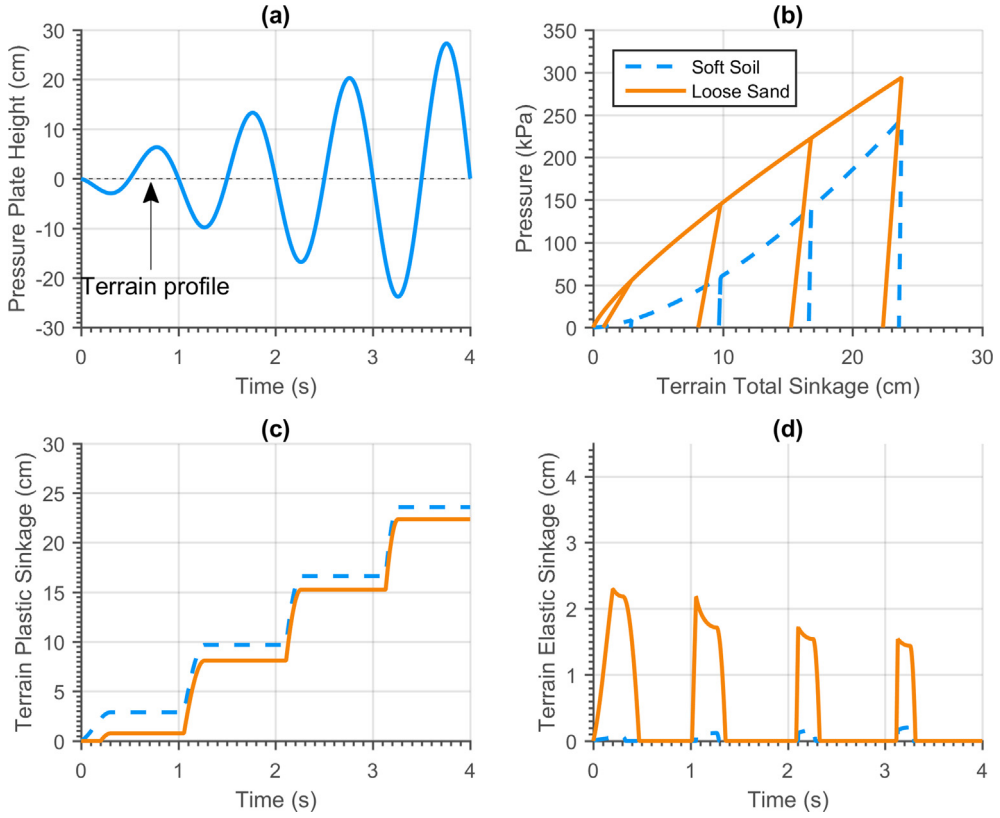


Fig. 10. Plate pressure loading of mineral terrains (a) pressure plate height, (b) terrain pressure vs. sinkage, (c) terrain plastic sinkage time history and (d) terrain elastic sinkage time history.

Table 1
Mechanical properties of mineral terrains used in simulations.

Terrain Type	Normal pressure			Repetitive loading		Shear stress (rubber-terrain)		
	n (−)	K_c $\left(\frac{kN}{m^{n+1}}\right)$	K_ϕ $\left(\frac{kN}{m^{n+2}}\right)$	K_o $\left(\frac{kN}{m^{n+1}}\right)$	A_u $\left(\frac{kN}{m^{n+1}}\right)$	C (kPa)	Φ (deg)	K (cm)
Soft soil	0.611	1.16	475.0	0	503,000	1.15	31.5	1.15
Loose sand	0.8	16.54	911.4	0	86,000	2.7	26.1	0.45

the previous sections. The developed models were implemented in the HSSTM package. During the simulation of the tire on non-deformable terrains, the terrain model provides the tire model with F&M at the center of the contact patch. On the other hand, on deformable terrains, the terrain model inputs the normal pressure and shear pressure distribution across the contact patch to the tire model. At the same time, the terrain model calculates the elastic and plastic terrain deformations, and updates the terrain profile respectively. In order to validate the performance of developed terrain models, a series of simulations were conducted on various terrain types, and with different simulation input conditions. To minimize the effect of tire structure compliance and focus on terrain model performance, the deformable terrain model was used with a rigid tire model. Also, the non-deformable tire model was used with the relaxation length model which is basically a simplified model for tire structure dynamics. The results are provided in the following sections, and discussions on the accuracy of the models are made.

6.1. Deformable terrain model

To evaluate the developed numerical terrain model, the pressure-sinkage characteristics of multiple terrain types were studied using a pressure plate loading simulation. During these tests, different flat terrain profiles were subjected to loading/unloading with a rectangular plate for which vertical position varied sinusoidally. The amplitude of the plate vertical displacement increased linearly with time. The pressure-sinkage characteristics for two mineral terrains during the aforementioned loading/unloading cycle are shown in Fig. 10. The mechanical terrain properties of mineral terrains that were used for conducting the simulations are described in Table 1 (Wong, 2008). The arbitrary terrains that are discussed in this section were only used to verify the terrain model capability by evaluating the respective terrain behaviors.

The overall terrain pressure-sinkage curve follows the trend that was discussed in Fig. 5. Every time the plate displacement goes below the initial flat terrain profile, the terrain pressure-sinkage follows the nonlinear loading path. Upon unloading the terrain, the pressure would decrease from the local maximum unloading pressure linearly to zero. The slope of the unloading/reloading line depends on the maximum unloading pressure, so it changes for each unloading cycle. The sinkage corresponding to the zero unloading pressure is the terrain plastic sinkage. During the reloading, when the loading plate goes below the plastic sinkage level from the previous unloading stage, the terrain pressure

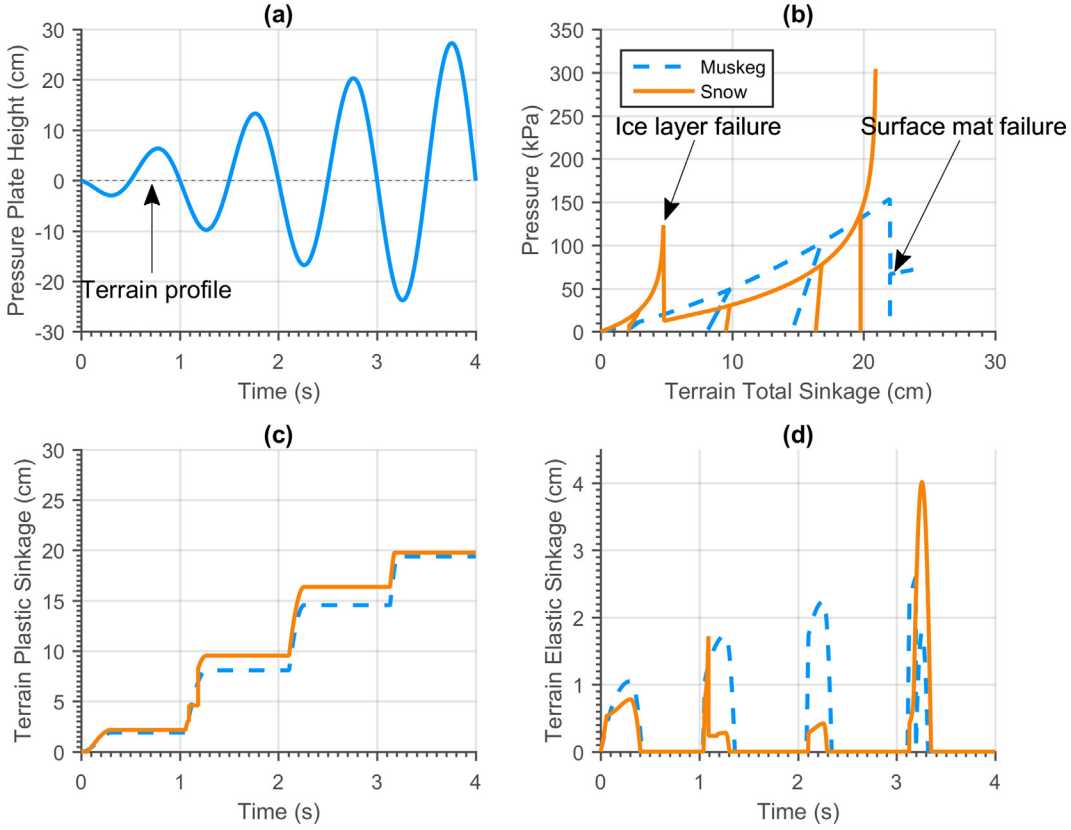


Fig. 11. Plate pressure loading of muskeg and snow (a) pressure plate height, (b) terrain pressure vs. sinkage, (c) terrain plastic sinkage time history and (d) terrain elastic sinkage time history.

Table 2
Mechanical properties of muskeg terrain used in simulations.

Terrain Type	Normal pressure				Repetitive loading		Shear stress (rubber-mat)		
	$k\ \left(\frac{kN}{m^3}\right)$	$m_m\ \left(\frac{kN}{m^3}\right)$	$z_b\ (cm)$	Peat Stiff.$\left(\frac{kN}{m^3}\right)$	$K_o\ \left(\frac{kN}{m^{n+1}}\right)$	$A_u\ \left(\frac{kN}{m^{n+1}}\right)$	C (kPa)	Φ (deg)	K (cm)
Petawawa Muskeg	338	53	22	302	334	25,430	1.7	40.3	0.9

Table 3
Mechanical properties of snow covered terrain.

Terrain type	Normal pressure				Repetitive loading		Shear stress (rubber-snow)		
	$k_{p1}\ (kPa)$	$k_{p2}\ \left(\frac{kPa}{cm}\right)$	$k_{z1}\ (cm)$	$k_{z1}\ (cm^2)$	$K_o\ \left(\frac{kN}{m^{n+1}}\right)$	$A_u\ \left(\frac{kN}{m^{n+1}}\right)$	C (kPa)	Φ (deg)	K (cm)
Petawawa Snow (before ice layer failure)	3.18	2.34	0.85	39.69	0	109,600	0.16	17	0.34
Petawawa Snow (after ice layer failure)	52.71	−0.48	14.21	67.34	0	109,600	0.16	16.1	0.35

increases through the unloading/reloading linear line until it reaches the local maximum unloading pressure from the last unloading step. Beyond this point, the pressure increase will follow the nonlinear loading path again until the next unloading step is initiated.

Furthermore, the formulation for non-mineral terrains was included in the terrain model as well. Fig. 11 shows the pressure-sinkage behavior of muskeg and snow terrains during the plate loading simulations. The parameters used for characterizing the models for these terrains are included in Tables 2 and 3. These values are derived from the literature (Bekker, 1956). The given experimental values are calculated based on pressure data for a 5 cm by 30 cm rectangular plate. Therefore, the width of the simulated tire was set equal to the width of the plate.

The process of loading the muskeg terrain is different than for the mineral terrains. The muskeg consists of a surface layer of fibrous composition and an underlying peat layer. The surface layer is idealized as a membrane-like structure that can sustain a certain level of sinkage. Once the sinkage passes this critical point, the surface mat will fail. The underlying peat is assumed to have a resistance proportional to its vertical deformation. As a result, once the surface mat layer fails, the pressure suddenly drops, and then increases linearly upon loading the terrain further with the loading plate. This behavior is correctly captured in simulation results shown in Fig. 11. It should be mentioned that the unloading/reloading characteristics of the muskeg is similar to that of mineral terrains.

During the loading of the snow-covered terrains, the terrain pressure increases gradually until the lower boundary of the deformation zone of the snow under the plate reaches the underlying ice layer. At this point, terrain pressure increases rapidly, and after a certain pressure level the ice layer fails. After fracture of the ice layer, further penetration of the plate into the snow beneath the ice layer causes the pressure to increase gradually again. This trend continues until the plate approaches the frozen ground at the base of the snow layer. Then, the pressure increases rapidly again, and the pressure-sinkage curve approaches an asymptotic line, as shown in Fig. 11(b).

The visualization of the tire-terrain interaction can help debug the implemented terrain model as well as offer a better insight into the physics of the problem. Thus, a visualization environment was implemented to illustrate the behavior of the user-defined terrain, such as the terrain plastic deformation and stress distribution. Fig. 12 shows snapshots of the terrain visualization environment for a tire crossing its previous path to visualize the terrain multi-pass behavior.

In this simulation, the tire starts traveling over the terrain in a straight line and it deforms the terrain surface, creating a rut. Next, the tire continues on a second path, which in this case is perpendicular to the first path. Because the HSSTM is a nonlinear system, the tire elements go under different states of normal and tangential stresses. As a result, the permanent plastic deformation of the ground after the tire passage is uneven. This deformation has a larger value at the crossing section of two paths, which has gone

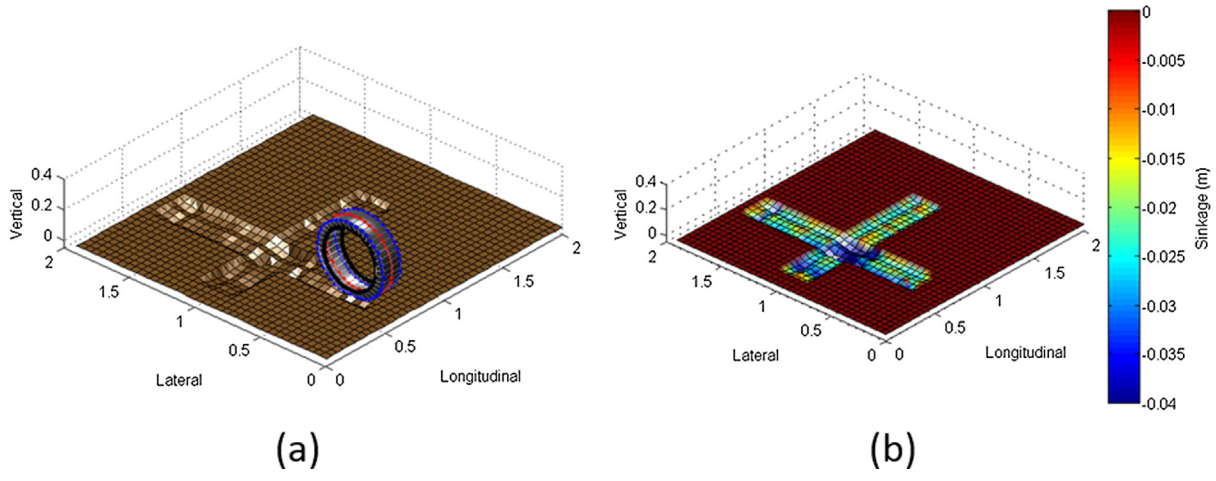


Fig. 12. Terrain visualization during the multi pass simulation of a tire with driving torque of 200 Nm and normal load of 4000 N: (a) soil consecutive loading-unloading and (b) terrain sinkage color contour.

Table 4
Mechanical properties of three mineral terrains used for simulations (Wong, 2010).

Soil Type	Bekker's equation			Moisture content (%)	Shear characteristics			
	$n(-)$	$k_c\ \left(\frac{kN}{m^{n+1}}\right)$	$k_\phi\ \left(\frac{kN}{m^{n+2}}\right)$		RS$\left(\frac{cm}{s}\right)$	c (kPa)	ϕ (deg)	K (cm)
Soft terrain (LETE soil)	0.611	1.16	475.0	0	2.5	1.15	31.5	1.15
Medium terrain (Upland sandy loam)	0.74	26.8	1522	44.3	2.5	2.7	26.1	4.2
Hard soil (Grenville loam)	1.01	0.06	5880	24.1	2.5	3.1	29.8	3.8

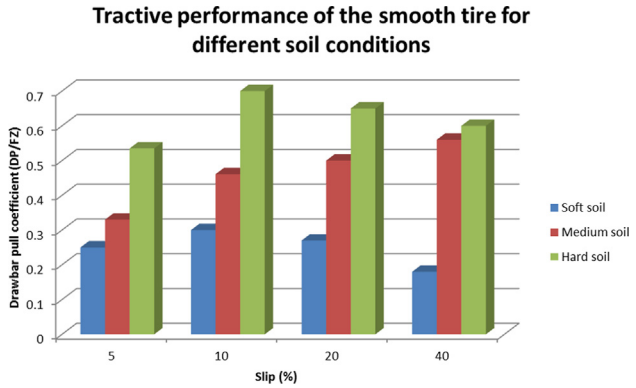


Fig. 13. Tractive performance of the buffed tire simulated at four slip ratios and three different soil conditions.

through deformation twice. The mechanical properties of the mineral terrain used for conducting this simulation are the same as the "Medium terrain" shown in Table 4. In this table n, k_c, and k_ϕ are the constants for the Bekker equation, RS is rate of shearing, c is soil cohesion, ϕ is the angle of shearing resistance, and K is the shear deformation parameter.

When the tire is moving on deformable terrain under an applied torque at its spindle, the positive shear forces keep pushing the tire forward, while the negative ground forces (rolling resistance, bulldozing force, etc.) resist the tire motion. The resultant force is called the drawbar pull, which is an indication for the ability of a vehicle to pull/push external load, accelerate, or overcome the grade resistance. Consequently, in calculation of the drawbar pull, both motion resistance due to tire flexing, soil compaction and slip are included. To normalize it, the drawbar pull is divided by the normal load at the spindle, thus obtaining the drawbar pull coefficient. The drawbar pull coefficient explicitly relates the tire tractive performance to the wheel slip ratio and implicitly to the terrain normal and shearing characteristics. Using the developed model, the drawbar pull coefficient was calculated at four slip ratio values and on three selective terrain types: soft, medium, and hard soil. The slip ratio i was calculated by normalizing the wheel slip velocity ${}_{wheel}V_{sx}$ with the carriage longitudinal velocity, ${}_{carriage}V_x$:

$$i = \frac{{}_{wheel}V_{sx}}{{}_{carriage}V_x} \times 100 = \frac{(R_{eff}\Omega - {}_{carriage}V_x)}{{}_{carriage}V_x} \times 100 \tag{41}$$

where R_{eff} is the wheel effective rolling radius and Ω is the wheel rotational velocity. The slip ratio values were maintained at their nominal values using a PID controller that regulated the applied torque to the spindle. The terrain mechanical properties used for the simulated terrains are documented in Table 4. The results of this simulation, which indicate the effect of the terrain properties on the mobility performance of the tire, are presented in Fig. 13.

To validate the normal pressure and shear stress distribution across the contact patch of a driven wheel, the performance of a rigid wheel with a diameter of 1245 mm and a width of 305 mm was simulated on the loose sand, and compared with the experimental data reported by Onafeko and Reece (1967). The simulation results for a vertical load of 9.28 kN, and two slip ratio values of 3.1% and 22.1% are reported in Fig. 14. The normal pressure and shear stress values are plotted against the polar coordinate (θ) starting beneath the tire center increasing in a counter clockwise direction.

The developed terrain model provides promising simulation results compared to experimental data. It can be seen that by increasing the tire slip ratio, the peak of the normal and shear stress curves shifts toward the front of the contact patch in the direction of moving. At the same time, the maximum value of the normal stress decreases while the shear stress increases. This is caused by the soil flow in the contact patch that increases as the tire travels with a higher slip ratio value. The vertical force at the wheel center is expressed by Wong (2010):

$$W = \frac{bD}{2}\left\{\int_0^{40}[\sigma(\theta)\cos\theta + \tau(\theta)\sin\theta]d\theta + \int_{-10}^{0}[\sigma(\theta)\cos\theta - \tau(\theta)\sin\theta]d\theta\right\} \tag{42}$$

where $b$is the tire width, D is the undeflected tire diameter, W is the normal load on the tire, σ is normal stress, and τ is shear stress. This equation was integrated over the contact length using the estimated and experimental stress distributions. The resulted normal load for simulation results is 9.28 kN that indicates wheel vertical equilibrium. The experimental normal load is slightly less (9.2 kN), which can be due to errors from digitizing the graphs.

6.2. Rigid terrain model

In order to find the required parameters for the rigid terrain model, the curve-fitted experimental data described by the Magic Formula (Rill, 2006; Onafeko and Reece, 1967) were used. The resulting data set is the smoothed version of the raw experimental data. This is due to the fact that Magic Formula consists of a set of empirical equations that captures the trend in the data in contrast to a physical model. The required Magic Formula parameters were adopted from the literature (Rill, 2006; Onafeko and Reece, 1967). Initially, the longitudinal and lateral force vs. slip curves were used to capture the normalized coulomb friction μ_c, normalized static friction μ_s, Stribeck relative velocity v_s, and steady-state friction force constant αused in the LuGre friction function:

$$g(v_r) = \mu_c + (\mu_s - \mu_c)e^{-|v_r/v_s|^\alpha} \tag{43}$$

The process of curve-fitting was conducted using a nonlinear least-square fitting method. The parameterized friction function $g(v_r)$ was incorporated in the developed distributed LuGre model to calculate the time dependent brush deformation in longitudinal and lateral directions. The parameterized terrain model was used along with the HSSTM tire model to conduct some cornering and braking/accelerating simulations. During the braking/accelerating tests, tire slip ratio was changed from −30% to 30% to −30% at different levels of normal load. The tire steady-state force generation characteristics as well its transient dynamic behavior were involved in this type of simulation. The experimental and

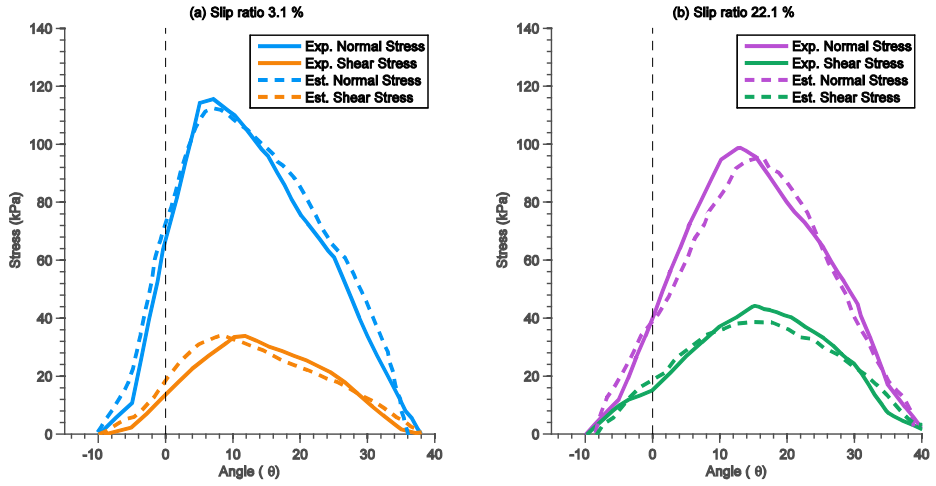


Fig. 14. Comparison of experimental and simulated normal and shear stress across the tire contact patch (a) slip ratio 3.1% and (b) slip ratio 22.1% (Experimental data from (Taheri and Wei, 2015).

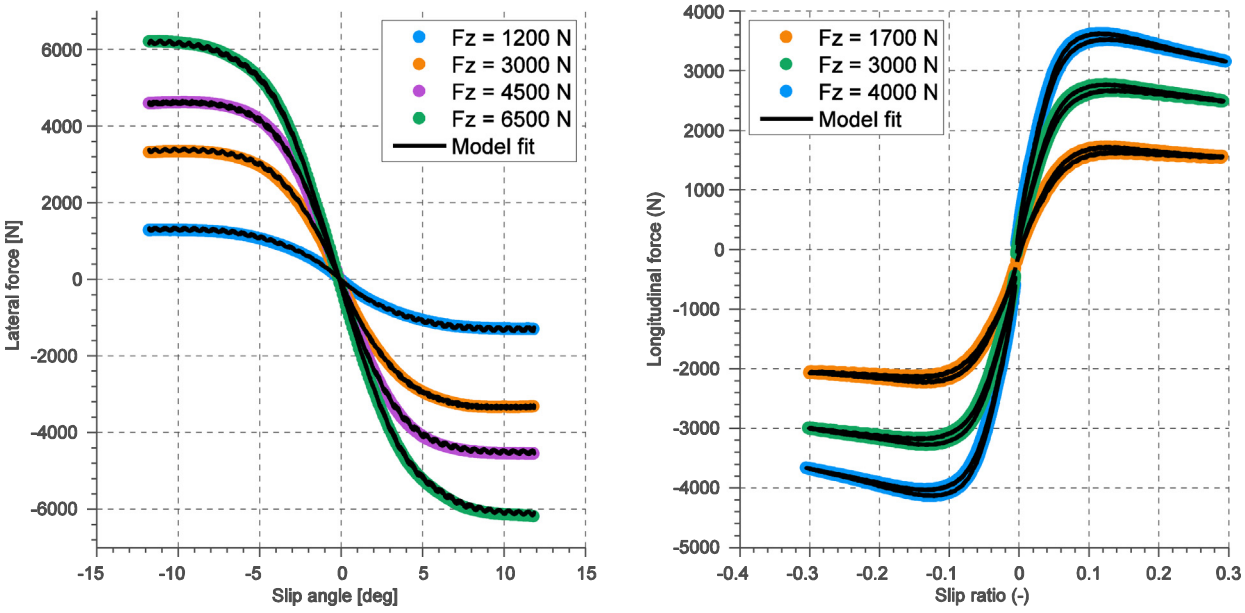


Fig. 15. Lateral and longitudinal forces during tire interaction with rigid terrain.

estimated tire longitudinal forces for Michelin LTX 235/85R16 are shown in Fig. 15. The longitudinal and lateral case studies are conducted on 120 grid sand paper, at tire nominal pressure and 80 km/h belt speed. As for the cornering case study, the tire slip angle was swept from −12 to 12 to −12 degree at different levels of normal load. The lateral force curves from experimental data as well as model estimations are shown in Fig. 15. The data used for identifying model parameters is shown with colored circles, and estimations at identical model input values using the parametrized brush model are shown with solid lines. As it was mentioned previously, the experimental data was regenerated using the Magic Formula (Pacejka and Bakker, 1992).

The distribution of lateral shear stress in the contact patch is not symmetric, which results in a nonzero torque around the z-axis known as aligning moment. The aligning moment tends to turn the tire about the z-axis and make it align with the direction of tire velocity vector. Similarly, the non-symmetric normal pressure distribution around x-axis produces a moment. The resultant overturning moment tends to turn the tire about the x-axis. The experimental and estimated values for the aligning moment and for the overturning moment are plotted in Fig. 16. It can be seen that the terrain model force and moment estimations are in good agreement with experimental results. The hysteresis loop in F&M curves is associated with the tire transient effect. This phenomenon causes the F&M curves to not follow the same trend during the increasing and decreasing slip value changes as can be seen in Figs. 15 and 16.

7. Conclusion

In this paper (Part II), the interaction modules used in conjunction with the Hybrid Soft Soil Tire Model (HSSTM) (described in Part I Sandu et al., 2019) were discussed. The terrain interaction module has contact search and contact interface units. For the contact interface unit, two computational models were developed for evaluating rigid and deformable terrain behaviors. The terrain model for deformable terrains is based on semi-empirical formula-

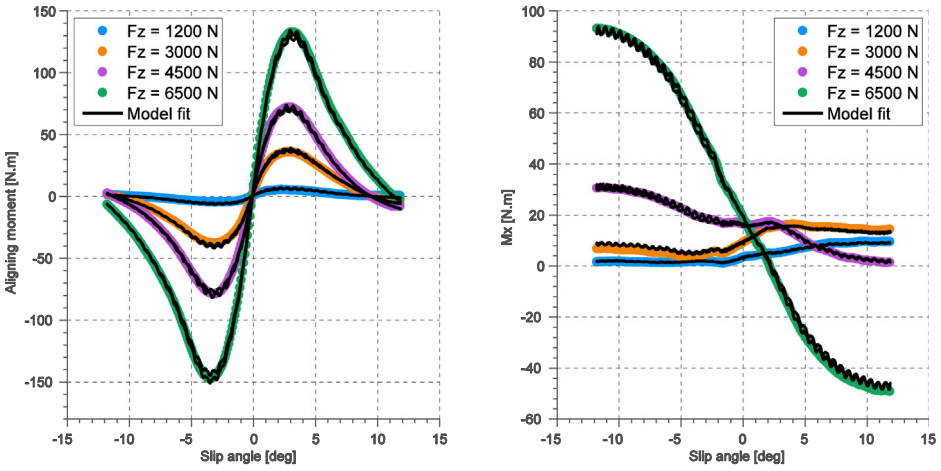


Fig. 16. Aligning and overturning moments during tire interaction with rigid terrain.

tions developed using the field test data for various types of terrains, such as mineral, muskeg, and snow covered terrains. Employing the empirical relations in addition to analytical approaches to describe terrain behavior helped achieve good results while keeping the computational effort and number of model variables low. However, this limits the domain of application to the conditions similar to those for which the test data was available. The semi-empirical formulations were implemented in a numerical scheme. The terrain profile model was implemented in RGR format and allowed fast tire-terrain contact detection. The terrain model capability in describing the pressure-sinkage and repetitive loading/unloading/reloading behavior of four different terrain types was illustrated.

A terrain visualization module was introduced to help debug the implemented terrain model, as well as to provide a better understanding of the physics of tire-terrain interaction. Furthermore, the normal pressure distribution and the shear stress distribution across the contact patch of a driven rigid wheel on loose sand were validated against experimental results at two different values of slip ratio. The rigid wheel was used for this purpose in order to focus on the terrain model accuracy.

The contact interface model for the rigid terrain model was based on a 3D distributed brush model enhanced with a LuGre friction model for dynamic maneuvers. The parameters for this rigid terrain brush contact model were obtained using a nonlinear least square method in order to minimize the error between the model force and moments estimations and the experimental data described by the Magic Formula (Pacejka and Bakker, 1992). The model developed was integrated with the HSSTM by connecting an array of massless, deformable bristles to each of the tire belt elements. Using this approach eliminates the need to assume a predefined normal pressure distribution over the tire contact patch.

The tire model parameterization and the HSSTM validation on different types of terrains is presented in Part III (Sandu et al.).

8. Disclaimer

Reference herein to any specific commercial company, product, process, or service by trade name, trademark, manufacturer, or otherwise, does not necessarily constitute or imply endorsement, recommendation, or favoring by the United States Government or the Department of the Army (DoA). The opinions of the author expressed herein do not necessarily state or reflect those of the United States Government or the DoA, and shall not be used for advertising or product endorsement purposes.

Acknowledgment

This project was funded in part by the Cooperative Agreement W56HZV-14-2-0001 US Army TARDEC/ARC.

References

Bekker, M.G., 1956. Theory of Land Locomotion: The Mechanics of Vehicle Mobility. The University of Michigan Press, Ann Arbor.
Bekker, M.G., 1960. Off the Road Locomotion. University of Michigan Press, Ann Arbor, Michigan.
Bekker, M.G., 1969. Introduction to Terrain-Vehicle System. The University of Michigan Press, Ann Arbor.
Chae, S., 2006. Nonlinear finite element modeling and analysis of a truck tire. In: Intercollege Graduate Program in Materials. Pennsylvania State University, p. 207.
Gallrein, A., Backer, M., 2007. CDTire: a tire model for comfort and durability applications. Veh. Syst. Dyn. 45 (1), 69–77.
Gipser, M., 2007. FTire - the tire simulation model for all applications related to vehicle dynamics. Veh. Syst. Dyn. 45, 139–151.
Janosi, Z., Hanamoto, B., 1961. Analytical determination of draw bar pull as a function of slip for tracked vehicles in deformable soils. In: Proceedings of the 1st International Conference of ISTVS, 1961. Turin, Italy.
Mao, S., Han, R.P., 2008. Nonlinear complementarity equations for modeling tire–soil interaction—An incremental Bekker approach. J. Sound Vib. 312 (3), 380–398.
Mastinu, G., Pairana, E., 1992. Parameter identification and validation of a pneumatic tyre model. Veh. Syst. Dyn. 21 (sup001), 58–81.
McCullough, M., Haug, E., 1986. Dynamics of high mobility track vehicles. J. Mech. Des. 108 (2), 189–196.
Nakashima, H., Oida, A., 2004. Algorithm and implementation of soil–tire contact analysis code based on dynamic FE–DE method. J. Terramech. 41 (2–3), 127–137.
Oertel, C., 1997. On modeling contact and friction calculation of tyre response on uneven roads. Veh. Syst. Dyn. 27 (S1), 289–302.
Oertel, C., Fandre, A., 1993. Ride comfort simulations and steps towards life time calculations: RMOD-K and ADAMS. In: International ADAMS User's Conference. Berlin, Germany.
Onafeko, O., Reece, A., 1967. Soil stresses and deformations beneath rigid wheels. J. Terramech. 4 (1), 59–80.
Pacejka, H., 2005. Tire and Vehicle Dynamics. Elsevier.
Pacejka, H.B., Bakker, E., 1992. The magic formula tyre model. Veh. Syst. Dyn. 21 (001), 1–18.
Rill, G., 2006. First order tire dynamics. In: III European Conference on Computational Mechanics Solids, Structures and Coupled Problems in Engineering, Lisbon, Portugal, Springer, The Netherlands.
Sandu, C., Taheri, Sh., Taheri, S., Gorsich, D., 2019. Hybrid Soft Soil Tire Model (HSSTM). Part I: Tire Material and Structure Modeling. J. Terramech. 86, 1–13.
Sandu, C., Taheri, Sh., Taheri, S., 2014. Development and Implementation of a Hybrid Soft Soil Tire Model (HSSTM). In: Proceedings of The 18th International Conference of the ISTVS, Sept. 22-25, 2014, Seoul, Korea.
Taheri, Sh., Sandu, C., Taheri, S., Naranjo, S., Jayakumar, P., Ross, B., Christ, D., 2013. Off-road soft soil tire model development, validation, and interface to commercial multibody dynamics software. In: The 7th Americas Regional Conference of the ISTVS, Nov. 4-7, 2013, Tampa, FL, USA.
Taheri, Sh., Sandu, C., Taheri, S., 2014. Finite element modeling of tire transient characteristics in dynamic maneuvers. SAE Int. J. Passeng. Cars - Mech. Syst. 7 (1), 221–230.

Taheri, Sh., Sandu, C., Taheri, S., Pinto, E., Gorsich, D., 2015. A technical survey on terramechanics models for tire-terrain interaction used in modeling and simulation of wheeled vehicles. J. Terramech. 57, p. 1-22. https://doi.org/10.1016/j.jterra.2014.08.003.

Taheri, S., Wei, T., 2015. A new semi-empirical method for estimating tire combined slip forces and moments during handling maneuvers. SAE Int. J. Passenger Cars-Mech. Syst. 8, 797–815.

Wong, J.Y., 2008. Theory of Ground Vehicles. John Wiley & Sons, Hoboken, NJ.

Wong, J.Y., 2010. Terramechanics and Off-Road Vehicle Engineering: Terrain Behaviour, Off-Road Vehicle Performance and Design. Elsevier, Amsterdam, The Netherlands.

Zavarise, G., De Lorenzis, L., 2009. The node-to-segment algorithm for 2D frictionless contact: Classical formulation and special cases. Comput. Methods Appl. Mech. Eng. 198 (41–44), 3428–3451.